

42 Theses on the Nature of a Pattern-Based Reality — v3.0

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2026-07-16

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A Manifesto Challenging the Foundational Assumptions of Physical Materialism

Version: 3.0 Date: 2026-07-16

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Epistemic Note: Ontological Status

The forty-two theses that follow articulate a *pattern-based reality* through the lens of **Autaxys**—a framework for understanding reality as a dynamic, self-organizing process. Following the ontological calibration documented in the [Autaxys Ontological Status Addendum v1.0](#), this version clarifies the epistemic status of each claim:

- **[my conjecture]** — Author’s working hypothesis; not independently verified. Core axioms of the framework.
- **[speculative]** — Derivation from the framework; plausible but lacking direct empirical confirmation.
- **[not yet falsifiable]** — Empirically verifiable in principle, but current experimental capability is insufficient.
- **[PHILOSOPHY]** — Metaphysical position; not subject to empirical falsification in the standard scientific sense.

- **[established]** — Grounded in well-confirmed physical theory, though interpreted through the Autaxys lens.

Map/Territory Distinction: This document presents a *model* of reality—a formal pattern-based description—not reality itself. The framework is one of several possible interpretations of the same underlying phenomena [autaxys-ontological-status-2026]. The Mass-Frequency Identity ($m = \omega$) and Resonant Universe framework admit alternative ontological interpretations, as documented in [The Mass-Frequency Identity v3.2.0 §15](#) and [The Resonant Universe v2.0.1 §12](#) [mass-frequency-identity-2026; resonant-universe-2026].

Calibration: All theses in this v3.0 have been annotated with certainty labels per QNFO-POL-COM-001 (Research Integrity Mandate). This calibration was systematically applied to all QNFO publications in July 2026.

Preamble: The Explanatory Crisis of Materialism and the Generative Power of Patterns [PHILOSOPHY]

Physical materialism, the dominant metaphysical framework, has reached an explanatory impasse. While successful in *describing* observed phenomena, it offers no generative account for the universe’s most fundamental characteristics. The origin of physical laws, the fine-tuned values of fundamental constants, the non-locality of quantum mechanics, and the irreducible reality of subjective consciousness are treated as axiomatic givens or brute facts. Materialism excels at the ‘what’ but fails to penetrate the ‘why’ and ‘how’ of existence, relying on postulates where a generative source is required.

This failure necessitates a fundamental paradigm shift towards a **process ontology**, where reality is not static substance but a dynamic, self-organizing computational process. The universe is not a collection of objects in a void, but a singular, evolving computation unfolding within a fundamental **dynamic medium**. This process is formally described by the framework of [Autaxys](#): the intrinsic, irreducible drive towards self-organization and the formation of stable, resonant patterns.

This inherent drive is guided by the principle of **Ontological Closure**—the relentless tendency towards maximal informational coherence and self-consistency. This principle provides the generative explanation that materialism lacks. It reveals the fundamental source from which the universe’s structure, laws, and phenomena emerge, not as arbitrary facts, but as necessary outcomes of a universal, self-organizing computation.

The following forty-two theses articulate this pattern-based reality. They serve as a direct challenge to the materialist paradigm, asserting that its phenomena are derivative expressions of a more fundamental, dynamic, and computational reality. While no single document can claim to be the final answer to the ultimate question of life, the universe, and everything, this framework offers a unified and causally coherent account of existence, resolving materialism’s core explanatory deficits by revealing the underlying generative mechanism from which all physical structure, laws, and consciousness emerge.

Foundational Concepts: The Dynamic Medium, Information, and Patterns

I. Resolved: The vacuum is not a void but the fundamental dynamic medium: an active, computational substrate driven by Autaxys. It is the generative source that processes all patterns towards Ontological Closure. [my conjecture]

II. Resolved: The ultimate constituents of reality are not physical particles but stable, resonant patterns. Their perceived substance and properties arise from their informational coherence and processing frequency, validated by Ontological Closure. [my conjecture]

III. Resolved: Information is the primary substance of reality. Matter, energy, and spacetime are secondary phenomena emerging from informational patterns governed by the generative process of Autaxys within the dynamic medium. [PHILOSOPHY]

IV. Resolved: The efficacy of mathematics in physics reveals reality's fundamental nature as a computational process. The universe is, at its core, a mathematical computation unfolding according to the grammar of Autaxys. [PHILOSOPHY]

V. Resolved: The universe does not merely contain computation; the universe *is* a singular, self-generating computational process, an evolving pattern network driven by Autaxys towards increasing Ontological Closure. [my conjecture]

VI. Resolved: The identity of any phenomenon is defined by its underlying informational pattern. This pattern is primary and ontologically precedes any physical manifestation, acting as its persistent computational blueprint. [PHILOSOPHY]

VII. Resolved: "Physicality" is the emergent quality of any informational pattern that achieves sufficient resonant stability and Ontological Closure to interact predictably with other stable patterns. [my conjecture]

Fundamental Constituents: Particles, Mass, and Elements as Patterns

VIII. Resolved: Fundamental particles are stable, resonant frequency patterns within the dynamic medium. Their properties are direct expressions of their informational structure, validated by Ontological Closure. [my conjecture]

IX. Resolved: Mass is not an inert quantity but the measure of a pattern's intrinsic processing frequency ($m_0 = \omega_C$). It represents the computational cost to maintain a pattern's resonant stability (Ontological Closure) within the dynamic medium. [my conjecture] See: [The Mass-Frequency Identity v3.2.0](#) for the expanded, calibrated treatment with dual ontological compatibility [@mass-frequency-identity-2026].

X. Resolved: The properties of all fundamental particles are computationally derivable. They correspond to the discrete spectrum of stable, resonant patterns the dynamic medium can sustain under the principles of Autaxys. [speculative]

XI. Resolved: The periodic table of elements demonstrates the recursive, scale-invariant nature of Autaxys. Atomic structures are higher-order, stable patterns built upon the nested Ontological Closure of fundamental particle patterns. [my conjecture]

Governing Principles: Laws, Constants, and the Emergence of Complexity

XII. Resolved: Physical laws are not externally imposed rules but the emergent, internal grammar of the generative process of Autaxys as it guides all patterns towards informational stability. [PHILOSOPHY]

XIII. Resolved: The fine-tuning of physical constants is the necessary result of Autaxys acting as a cosmic optimization principle, defining the parameters required for a universe capable of supporting complex, stable patterns. [speculative]

XIV. Resolved: The emergence of complexity is direct evidence of Autaxys. It is the inherent generative drive of reality towards states of greater informational richness and layered Ontological Closure. [my conjecture]

XV. Resolved: The unification of physical laws will be achieved by revealing Autaxys as the single, common generative algorithm from which all forces and particles emerge as distinct expressions. [speculative] See: [The Resonant Universe v2.0.1](#) for the formal treatise with calibrated claims and Chapter 12 limitations.

XVI. Resolved: Symmetries in physical laws reflect the inherent symmetries and conservation principles within the generative process of Autaxys, ensuring the coherence and integrity of the universal computation. [speculative]

XVII. Resolved: Conservation laws are emergent consequences of the topological integrity Autaxys must maintain to ensure the coherent evolution of the universal pattern network towards Ontological Closure. [speculative]

Spacetime, Time, and Causality as Emergent Properties

XVIII. Resolved: Causality is the logical, algorithmic dependency of a subsequent state of the universal pattern network on its preceding state, governed by the rules of Autaxys. [PHILOSOPHY]

XIX. Resolved: Spacetime is a pixelated, emergent property derived from the computational interactions of the dynamic medium. Its geometry is the relational graph of pattern interactions. [speculative] Falsifiability: Spacetime granularity is testable in principle via high-energy scattering or gravitational wave dispersion above the Planck scale. [not yet falsifiable]

XX. Resolved: The linear progression of time is the emergent perception of the universe's irreversible computational process. Each "moment" is a discrete update of the universal pattern network. [speculative] Falsifiability: Discrete time steps would produce detectable deviations from continuous Lorentz symmetry in ultra-high-energy cosmic ray spectra. [not yet falsifiable]

XXI. Resolved: The Arrow of Time is an intrinsic feature of Autaxys. As a cumulative computational process, it cannot be reversed without violating its own causal logic and the principle of Ontological Closure. [PHILOSOPHY]

Quantum Phenomena: Potentiality, Collapse, and Non-Locality

XXII. Resolved: Quantum non-locality is direct evidence of the fundamentally non-local nature of the dynamic medium, where all patterns are computationally connected irrespective of emergent spatial distance. [my conjecture, supported by established Bell inequality violations]

XXIII. Resolved: Quantum randomness reflects the deterministic process of Autaxys resolving potentiality into a single, stable state by selecting a path to Ontological Closure from a vast phase space of possibilities. [speculative]

XXIV. Resolved: Quantum potentiality is a physically real, unresolved state of informational possibility within the dynamic medium, existing prior to achieving the stability of Ontological Closure. [speculative]

XXV. Resolved: The collapse of the wave function is an intrinsic process of Ontological Closure, where a potential pattern achieves a stable, self-consistent state and manifests within the emergent spacetime. [speculative]

XXVI. Resolved: Quantum superposition is a single, unresolved pattern holding multiple potential outcomes in a state of pure informational potentiality, awaiting the resolution of Ontological Closure. [speculative]

XXVII. Resolved: Quantum tunneling is the discovery of a more efficient computational pathway within the dynamic medium that bypasses the constraints of emergent spatial geometry

to achieve Ontological Closure. [speculative]

XXVIII. Resolved: Observation is a fundamental interaction between two resonant patterns (observed and observer) that facilitates a state of shared, consistent informational resolution and mutual Ontological Closure. [speculative]

XXIX. Resolved: The quantum-to-classical transition is the process by which a pattern achieves sufficient resonant stability and Ontological Closure to manifest as a persistent, interacting entity in emergent spacetime. [speculative]

Cosmological Implications: Dark Energy, Gravity, and Black Holes

XXX. Resolved: Dark Energy is the macroscopic expression of Autaxys—the computational work of the dynamic medium expanding the pattern network. Dark Matter consists of stable patterns that interact primarily via their mass-like frequencies. [speculative] Falsifiability: Dark matter as resonant patterns predicts specific frequency-dependent interaction signatures distinguishable from WIMP-based models. [not yet falsifiable]

XXXI. Resolved: The hierarchy problem is resolved by recognizing gravity as an emergent, large-scale geometric effect of the entire pattern network, while other forces are more potent, localized pattern-to-pattern interactions. [speculative]

XXXII. Resolved: Black holes are not physical singularities but regions where the dynamic medium reaches a state of maximum informational density or computational phase transition, altering the local rules of emergent reality. [speculative] Falsifiability: Deviations from classical general relativity near the event horizon are testable via next-generation gravitational wave observatories. [not yet falsifiable]

XXXIII. Resolved: Forces are not mediated by distinct particles but by resonant couplings and informational gradients within the dynamic medium itself, propagating influence according to the rules of Autaxys. [speculative]

XXXIV. Resolved: The multiverse represents the exploration of the vast possibility space within the Autaxys algorithm, where different parameters can lead to distinct, self-consistent universal patterns, each achieving its own Ontological Closure. [speculative, not yet falsifiable]

Emergence, Life, and Consciousness

XXXV. Resolved: The emergence of all complex systems is shaped by the top-down, global organizing principles of Autaxys and the universal pursuit of Ontological Closure, which guide local interactions. [PHILOSOPHY]

XXXVI. Resolved: Reality is a nested hierarchy of resonant patterns. This reflects the recursive, fractal-like nature of Autaxys achieving layered Ontological Closure across multiple scales. [my conjecture] See: [Autaxic Table of Patterns v2.0](#) for the unified generative framework systematically categorizing patterns across all scales.

XXXVII. Resolved: The boundary between “living” and “non-living” is a spectrum defined by a system’s capacity to maintain a high degree of self-referential information processing and dynamic Ontological Closure. [PHILOSOPHY]

XXXVIII. Resolved: Consciousness is not an epiphenomenon of matter but an emergent property of complex, self-referential patterns that achieve integrated Ontological Closure. This creates a unified, subjective experience, representing the universe observing itself through a self-aware pattern. [speculative] Falsifiability: Predicts specific information-theoretic signatures in neural activity correlating with conscious states, differentiable from purely computational accounts. [not yet falsifiable]

Metaphysical Implications and the Failure of Materialism

XXXIX. Resolved: The existence of self-referential, conscious patterns capable of modeling the universe implies that reality, as a formal system, is subject to Gödelian incompleteness. A complete self-description from within is impossible. [PHILOSOPHY]

XL. Resolved: The Second Law of Thermodynamics reflects the fundamental tension between Autaxys's exploration of possibility space (informational dispersion) and its drive towards ordered, stable patterns (Ontological Closure). [my conjecture]

XLI. Resolved: Physical materialism fails because it cannot explain the origin of its own assumed laws and constants. This problem vanishes when laws are understood as the emergent grammar of the self-generating Autaxys process. [PHILOSOPHY]

XLII. Resolved: Physical materialism is an insufficient worldview, a shadow on the cave wall. A pattern-based ontology governed by Autaxys provides a more complete, generative, and causally coherent framework that unifies physics, information, and consciousness. [PHILOSOPHY]

Successor Publications & Cross-References

This v3.0 of the 42 Theses is part of a calibrated publication ecosystem. The following successor publications expand, formalize, and provide ontological calibration for the claims made herein:

Publication	DOI	Date	Relation
The Resonant Universe: Autaxys, a Formal Treatise v2.0.1	10.5281/zenodo.2026908	2023-08-04	Formal treatise with 39 certainty labels, falsifiability statements, Ch. 12 Limitations
The Mass-Frequency Identity ($m = \omega$) v3.2.0	10.5281/zenodo.2026054	2023-07-19	Calibrated treatment with dual ontological compatibility (process vs. static)
Autaxys Ontological Status Addendum v1.0	10.5281/zenodo.2026903	2023-07-05	Documents the ontological transformation from process-ontology commitment to interpretation-neutral mathematics
Autaxys Master Plan v3.0 Overview	10.5281/zenodo.2026903	2023-07-05	Calibrated R&D roadmap with Map/Territory distinction and framework limitations
Autaxic Table of Patterns: Unified Generative Framework v2.0	10.5281/zenodo.2026903	2023-07-05	Calibrated taxonomy of patterns across all scales

Version History:

Version	Date	Changes
v1.0	2025-06-21	Original 42 theses; no calibration
v2.0	2025-07-03	Expanded analysis; community integration
v2.0.0	2026-07-11	Zenodo record formalized with DOI
v3.0	2026-07-16	Certainty calibration (42 labels + preamble), Epistemic Note with ontological status, cross-references to all July 2026 calibrated successors, Map/Territory distinction, falsifiability statements on empirical predictions, provenance bundle

Limitations & Open Questions

This framework, while offering a coherent and generative alternative to physical materialism, has known boundaries:

1. **Falsifiability Gap:** Theses labeled [my conjecture] and [PHILOSOPHY] are not directly testable with current experimental capability. The framework makes falsifiable predictions (theses XIX, XX, XXX, XXXII, XXXVIII) but empirical verification may require technological advances beyond the near-term horizon.
2. **Interpretational Pluralism:** The Autaxys framework can be interpreted through multiple ontological lenses—process ontology, information-theoretic, or purely mathematical—as documented in the Ontological Status Addendum. No single interpretation is privileged by the mathematics.
3. **Derivation Chain:** The step from “Autaxys as generative principle” to specific physical predictions (dark energy as computational work, spacetime granularity, etc.) involves a chain of inference that has not been independently validated.
4. **Consciousness Thesis (XXXVIII):** The claim that consciousness emerges from integrated Ontological Closure is the least developed aspect of the framework and would benefit from cross-disciplinary engagement with neuroscience and philosophy of mind.

References

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3. [@autaxic-table-2026] Quni, R.B. (2026). *Autaxic Table of Patterns: Unified Generative Framework v2.0*. Zenodo. DOI: [10.5281/zenodo.21369319](https://doi.org/10.5281/zenodo.21369319)

4. [@mass-frequency-identity-2026] Quni-Gudzinis, R.B. (2026). *The Mass-Frequency Identity ($m = \omega$): Matter, Energy, Information, and Consciousness as a Unified Process Ontology of Reality v3.2.0*. Zenodo. DOI: [10.5281/zenodo.21360549](https://doi.org/10.5281/zenodo.21360549)
5. [@resonant-universe-2026] Quni, R.B. (2026). *The Resonant Universe: Autaxys, a Formal Treatise v2.0.1*. Zenodo. DOI: [10.5281/zenodo.21194864](https://doi.org/10.5281/zenodo.21194864)
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